

Speech Processing Quiz 2

CSC325 · Speech Processing

TOTAL MARKS 20	DURATION 30 min	DIFFICULTY MEDIUM	THRESHOLD 75%	QUESTIONS ?
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STUDENT NAME	STUDENT ID	SECTION	DATE
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INSTRUCTIONS

- Read all questions carefully before answering.
- All questions are compulsory unless stated otherwise.
- Total time allowed: 30 minutes.
- No electronic devices allowed during the quiz.
- Write clearly and legibly in the space provided.

Questions

Assessment Questions

- 1 In the context of machine learning, when dealing with environments that have complex and unpredictable dynamics, which type of algorithms are most suitable?

[1 mark | Applying | medium]

- ☐ A Reinforcement Learning
- ☐ B Linear Regression
- ☐ C K-Means Clustering
- ☐ D Principal Component Analysis

Student Answer: &

- 2 Explain what windowing is in digital signal processing and why it is important for FFT analysis.

[5 marks | Understanding | medium]

Answer:

- 3 The Nyquist sampling theorem states that the sampling rate must be at least _____ times the highest frequency component in the signal.

[3 marks | Remembering | medium]

The Nyquist sampling theorem states that the sampling rate must be at least _____ times the highest frequency component in the signal.

1 blank(s) to fill:

- 4 True or False: A finite impulse response (FIR) filter can have non-linear phase response.

[2 marks | Understanding | medium]

- ☐ A. True
- ☐ B. False

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Provide a detailed explanation of spectral leakage in FFT and discuss the implications for speech signal analysis.

[9 marks | Analyzing | medium]

Answer:
